

Mohamed Adel Samy

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Nationality: Egyptian · Iqama: Valid · Date of birth: 09/02/1996

PROFESSIONAL SUMMARY

Senior Software Engineer with 7+ years delivering and operating web systems for government, utility and enterprise clients across Saudi Arabia, Oman and Egypt. Works end to end: React and TypeScript front ends, NestJS, Node.js and Java services in an event-driven microservices architecture, and the on-premise infrastructure underneath them. Currently sole engineer on a national environmental monitoring platform for Saudi Arabia's National Center for Environmental Compliance, covering 200+ sensors across the Kingdom, deployed highly available on RHEL with Docker Swarm and a PostgreSQL Patroni cluster. Track record of measurable delivery: 40% faster API responses, 30% less downtime, 50% less manual processing.

TECHNICAL SKILLS

Languages TypeScript, JavaScript (ES6+), Java, SQL, HTML, CSS

Frontend React, Next.js, Redux, Chakra UI, Tailwind CSS

Backend NestJS, Node.js, Java (Spring Boot), REST APIs, Microservices

Event-Driven & Streaming Apache Kafka, RabbitMQ (HA clustering), Debezium (change data capture)

Databases PostgreSQL, PostgreSQL Patroni (HA, automatic failover), PostGIS, Redis, query tuning, backup and recovery

Infrastructure Docker, Docker Swarm, NGINX, on-premise high-availability deployment, firewall configuration, Azure (via Azure CLI), GitHub Actions

Linux RHEL, Ubuntu, server provisioning and administration, network and static IP configuration

AI Engineering LLM integration, schema-constrained structured output, response validation and repair, automated evals, Cloudflare Workers

Integration & Workflow WSO2 Micro Integrator (low-code integration services), Camunda BPM (BPMN, DMN, CMMN), REST and message-based system integration, IoT platform integration

GIS & Geospatial Maps & Apps (MnA), PostGIS, Apache Superset, ERDAS IMAGINE

Practices Agile, code review, automated testing, AI-assisted development (Claude Code, Antigravity), WCAG 2.2 accessibility, Core Web Vitals

PROFESSIONAL EXPERIENCE

Senior Software Engineer Penta-b · January 2022 to present

- Sole engineer on the NCEC environmental compliance platform (Saudi Arabia): owned MnA configuration, high-availability deployment, real-time sensor integration and compliance plugin development end to end. See Selected Projects.
- Deploy and operate production stacks on RHEL with Docker and Docker Swarm behind NGINX, including PostgreSQL Patroni clusters for automatic database failover and hardened firewall configuration, reducing system downtime by 30%.
- Build event-driven microservices on the MnA platform using Apache Kafka, with Debezium capturing change events from selected PostgreSQL tables so downstream services stay consistent with the source of record.
- Build low-code integration services with WSO2 Micro Integrator, and deployed and operate a highly available on-premise RabbitMQ cluster carrying live sensor telemetry between client systems and the platform.
- Designed REST APIs and a microservices split that cut response times by 40%, introducing Redis caching on read-heavy paths.
- Automated licensing and approval workflows with Camunda BPM using BPMN and DMN models, removing manual routing between departments and cutting processing effort by 50%.
- Led backend integration of smart asset tracking across GIS and IoT platforms for two smart-city developments.
- Deliver React and TypeScript front ends against NestJS and Java services, deploying to both Azure infrastructure via the Azure CLI and on-premise hosts, working directly with clients to turn requirements into shipped systems.

GIS Developer Edge-Pro for Information Systems · July 2020 to January 2022

- Built and customised Mapp Enterprise dashboards with its JavaScript API, improving data visualisation and user interaction for client reporting.
- Designed form-management and notification workflows, improving cross-system communication and reporting turnaround.
- Ran remote sensing and satellite image analysis in ERDAS IMAGINE, including water pollution detection for environmental research.
- Delivered pre-sales support and client technical training on Skyline and ERDAS IMAGINE, plus internal documentation and knowledge-sharing sessions.

Full-Stack Developer Freelance · January 2019 to June 2020

- Delivered React dashboards and full-stack applications for small businesses, automating internal workflows.
- Built and deployed REST APIs on NestJS and PostgreSQL, tuned for the performance and scale each client actually needed.

SELECTED PROJECTS

Environmental Compliance Monitoring — National Center for Environmental Compliance (NCEC), Saudi Arabia · current

Sole engineer on a national platform monitoring noise and light from **200+ sensors deployed across the Kingdom**, built on Penta-b's Maps & Apps (MnA). Responsible for the entire delivery: platform configuration, infrastructure, integration and application code.

- Deployed the full stack for high availability on RHEL: Docker Swarm orchestration, a PostgreSQL Patroni cluster giving automatic failover, NGINX, and firewall configuration to meet the client's security requirements.
- Built the backend real-time integration service ingesting live readings from the sensor estate over a highly available on-premise RabbitMQ cluster, with WSO2 Micro Integrator mediating between the sensor vendors' systems and the platform.
- Developed the NCEC compliance frontend plugins on MnA: live dashboards, geospatial views of the sensor network, and a configurable alarming system that raises alerts when readings breach regulatory thresholds.

React · TypeScript · Node.js · MnA · WSO2 Micro Integrator · RabbitMQ · PostgreSQL Patroni · Docker Swarm · RHEL

Water and Wastewater Resource Management — Cairo, Giza and Alexandria Governorates

Led design and deployment of a resource management platform running across three governorates, giving thousands of users real-time issue tracking and resolution against field infrastructure.

React · Java · Camunda BPM · Docker

Licensing and Public Complaint Workflows — Alexandria Governorate

Camunda-modelled processes for licence applications and public complaints, replacing manual routing between departments. Routing rules and fee thresholds live in the process model rather than application code, so policy changes ship without a deployment. **Cut manual approvals by 50%.**

Camunda BPM · Java · PostgreSQL · Docker

Geo-Enabled E-Services — Ministry of Tourism, Oman

Geo-aware features for the ministry's public tourism portal, integrating GIS mapping and REST APIs to streamline visa applications and tourist site registrations.

React · PostGIS · Docker Swarm

Smart City Asset Management — Emaar Egypt City and Olympic City

Asset management and maintenance tracking for two developments, integrating GIS and IoT platforms behind REST services with automated flows between the systems feeding them.

React · Java · NestJS · n8n · Docker Swarm · NGINX

EDUCATION

Master of Business Administration — Brooklyn Business School · 2024 to present

Full Stack Web Development Diploma — Route Academy · 2021 to 2022

BSc Civil Engineering — German University in Cairo (GUC) · 2013 to 2018

LANGUAGES

Arabic (native) · English (professional) · German (basic)